On the Guidance of Flow Matching: review on synthetic data

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Abstract

Flow matching has shown state-of-the-art performance in various generative tasks in image generation and scientific applicaiton. In many applicatons, it is desirable to generate specific parts of the target distribution which can be dynmaically controlled after training. Guidance methods have been proposed to steer a frozen flow matching model towards a defined objective. In this paper, we evaluate the guidance methods for flow matching on synthetic data and discuss the limiting assumptions of the methods.

1 Introduction

Deep generative models have achieved remarkable success across a wide range of domains, from high-fidelity image synthesis (Rombach et al., 2022) to computational biology (Klein et al., 2025) and drug discovery (Jumper et al., 2021). A shared principle behind these successes is the idea of learning a mapping from a simple, tractable distribution (e.g., standard Gaussian) to the complex target distribution of interest, enabling the model to generate realistic samples by transforming random noise. Diffusion models (Ho et al., 2020; Song et al., 2021; Rombach et al., 2022) realize this by learning to reverse a gradual noise-corruption process, and have set strong baselines across many benchmarks in computer vision. Flow matching (Lipman et al., 2023) approaches the same goal more directly. A time-dependent vector field is defined to continuously deform the prior distribution into the target data distribution, and a neural network is trained to approximate this field from data. At inference, new samples are generated by constructing a trajectory from a point in the source distribution to the target distribution through the learned vector field. This can be achieved by numerically integrating an ordinary differential equation (ODE). A critical design choice is the coupling, that is, how individual source samples from the prior are paired with target samples from the data distribution when constructing the training trajectories. Randomly coupling source and target points produces crossing trajectories that place conflicting demands on the vector field and make learning harder, whereas pairing via optimal transport yields shorter, straighter paths that are easier to approximate and require fewer ODE integration steps at inference (Tong et al., 2024). Entropic variants of this coupling have further extended flow matching to unpaired data settings such as single-cell genomics (Klein et al., 2024), broadening its practical applicability.

In many applications it is desirable to generate not just any sample from the target distribution, but samples that satisfy a user-specified criterion, for instance, images of a particular class. One natural approach is to incorporate this criterion at training time, for example by conditioning the model on a class label or other side information (Ho and Salimans, 2022); the resulting model can then generate samples from the desired subdistribution on demand. However, such training-time conditioning must be built into the model from the start, and obtaining samples under a new or different objective requires retraining.

A more flexible alternative is inference-time steering, commonly referred to as guidance. There, the generation process of an already trained model is modified at inference time to bias sampling toward

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the desired region of the target distribution, without any retraining (Dhariwal and Nichol, 2021; Feng et al., 2025). For diffusion models, this problem is well understood. Classifier guidance (Dhariwal and Nichol, 2021) and classifier-free guidance (Ho and Salimans, 2022) modify the score function at inference time to bias generation toward a desired condition. Transferring these ideas to flow matching is, however, non-trivial, because the deterministic trajectories and the absence of a stochastic noise schedule fundamentally change how an energy function can be incorporated into the velocity field. As a consequence, guidance for flow matching has remained largely underexplored compared to its diffusion counterpart. Feng et al. (2025) recently proposed the first general framework for guidance in flow matching, deriving a family of training-free and training-based techniques covering approximate and asymptotically exact methods. However, a systematic empirical comparison of these approaches and a characterization of their failure modes has not been carried out.

We summarize our contributions as follows:

- We provide a review of the guidance methods for flow matching.
- We experimentally test the guidance methods on synthetic data, visualize their behavior, and compare their performance in terms of different metrics.
- We experimentally examine the limitations of the guidance methods regarding different coupling strengths and desired subdistributions and their effect on the generated samples.

2 Background

2.1 Conditional Flow Matching

Flow Matching (FM) is a generative method that learns a time-dependent velocity field $v_t(x)$, $t \in [0,1]$, $x \in \mathbb{R}^d$, that transports a tractable source distribution p_0 (e.g. a standard Gaussian) into a complex target distribution p_1 on $\mathbb{R}^d$ (Lipman et al., 2023). In particular, the velocity field is defined as $v_t(x) : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$, i.e., the source and target distributions are on the same space. Starting from $x_0 \sim p_0$, the velocity field defines a trajectory x_t through the ordinary differential equation (ODE)

$$\frac{d}{dt} x_t = v_t(x_t), \qquad x_t\big|_{t=0} = x_0. \tag{1}$$

Following this ODE over time from $t = 0$ to $t = 1$ continuously deforms the source into the target distribution. The distribution of x_t, denoted p_t, coincides with p_0 at $t = 0$ and with p_1 at $t = 1$, i.e. $x_t\big|_{t=1} = x_1 \sim p_1$. At inference time, a new sample is generated by drawing $x_0 \sim p_0$ and numerically integrating Eq. (1) forward to $t = 1$.

Ideally, v_t would be trained by regressing it onto the true marginal vector field $u_t(x)$ that generates the path p_t. Conditioning on a latent variable z, e.g. an endpoint pair $z = (x_0, x_1)$ with $x_0 \sim p_0$, $x_1 \sim p_1$, for which a conditional probability path $p_t(x|z)$ and a conditional velocity field $u_t(x|z)$ are chosen in closed form, the marginal path is recovered as $p_t(x) = \int p_t(x|z)\, p(z)\, dz$, and the marginal vector field can be written as a conditional expectation of $u_t(x|z)$ over the posterior $p(z|x_t = x)$ (Lipman et al., 2023):

$$u_t(x) = \int u_t(x|z)\, \frac{p_t(x|z)\, p(z)}{p_t(x)}\, dz. \tag{2}$$

Training v_t to match this marginal target directly amounts to minimizing

$$\mathcal{L}_{FM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, x_t\sim p_t} \left[\|v_t(x_t) - u_t(x_t)\|^2 \right]. \tag{3}$$

This expectation is intractable in practice because evaluating $u_t(x_t)$ requires the integral in Eq. (2), and sampling $x_t \sim p_t$ requires already knowing the path we are trying to learn. Lipman et al. (2023) show that Eq. (3) can be replaced by a tractable conditional counterpart: regressing v_t onto the closed-form conditional target $u_t(x|z)$ yields the same gradient with respect to v_t as regressing onto the intractable marginal target $u_t(x)$. This gives the simulation-free Conditional Flow Matching (CFM) loss

$$\mathcal{L}_{CFM} = \mathbb{E}_{t\sim\mathcal{U}(0,1),\, z\sim p(z),\, x_t\sim p_t(\cdot|z)} \left[\|v_t(x_t) - u_t(x_t|z)\|^2 \right], \tag{4}$$

where $u_t(x_t|z)$ is the conditional velocity field that transports x_0 to x_1 along the path defined by $z = (x_0, x_1)$. Crucially, every variable in Eq. (4) is obtained without ever simulating the ODE in Eq. (1): t is drawn uniformly from $[0,1]$; $z = (x_0, x_1)$ is obtained by sampling $x_0 \sim p_0$ and $x_1 \sim p_1$ from the training data, or jointly from a coupling (Section 2.2). Moreover, there are infinitely many

ways to define a trajectory from x_0 to x_1. Section 2.2 explains different choices which are simple to compute in closed form. Furthermore, $v_t(x_t)$ can simply be computed with a forward pass of the neural network being trained.

In the case of the linear, "optimal transport" (OT) conditional path, the quantitis x_t and $u_t(x_t|z)$ can be computed in closed form from t and z:

$$x_t = (1-t)\,x_0 + t\,x_1, \qquad u_t(x_t|z) = x_1 - x_0 = \frac{x_1 - x_t}{1-t}, \tag{5}$$

where the conditional probability path moves x_0 to x_1 along a straight line at constant velocity. The second, equivalent form of u_t in Eq. (5) reappears directly in the velocity corrections used for guidance in Section 3.

2.2 Coupling

Equation (5) shows that the conditional path is fully determined once the endpoint pair $z = (x_0, x_1)$ is fixed; how source and target samples are paired to form z is therefore itself a design choice, called the *coupling* π. Formally, a coupling is any joint distribution $\pi(x_0, x_1) \in \Pi(p_0, p_1)$ with marginals p_0 and p_1, and $z \sim \pi$ replaces the independent sampling $x_0 \sim p_0,\ x_1 \sim p_1$ used above. The simplest choice is the *independent coupling* $\pi = p_0 \otimes p_1$, which pairs source and target points at random (Independent-CFM). As shown in Fig. 1, this can pair nearby source points with distant, unrelated target points, producing trajectories that cross one another and place conflicting demands on v_t at the same (t, x), making the vector field harder to learn.

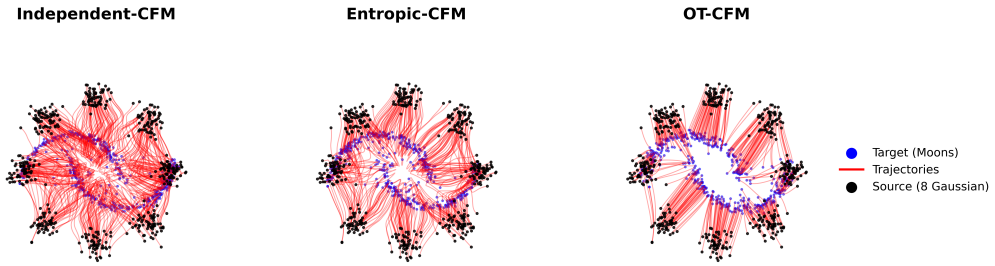


Figure 1: Comparison of three coupling strategies for flow matching: Independent-CFM, Entropic-CFM, and OT-CFM. Each panel shows sample trajectories (red) from the source distribution (8 Gaussians, black) to the target distribution (Moons, blue). OT-CFM produces shorter, straighter trajectories by minimizing transport cost, while Independent-CFM yields crossing paths.

The *optimal transport (OT) coupling* instead chooses π to minimize the expected squared transport cost,

$$\pi^\star = \underset{\pi \in \Pi(p_0,p_1)}{\arg\min}\ \mathbb{E}_{(x_0,x_1)\sim\pi}\left[\|x_1 - x_0\|^2\right], \tag{6}$$

the Kantorovich formulation of optimal transport. Sampling $z \sim \pi^\star$ yields the shorter, non-crossing trajectories seen in the OT-CFM panel of Fig. 1, which are easier for v_t to approximate and require fewer ODE integration steps at inference (Tong et al., 2024).

Exact OT couplings are computed per minibatch and can be costly and unstable to estimate for empirical distributions. The *entropic coupling* regularizes Eq. (6) with an entropy term,

$$\pi^\star_\epsilon = \underset{\pi \in \Pi(p_0,p_1)}{\arg\min}\ \mathbb{E}_{(x_0,x_1)\sim\pi}\left[\|x_1 - x_0\|^2\right] - \epsilon\, H(\pi), \tag{7}$$

which can be solved efficiently with the Sinkhorn algorithm. The regularization strength ϵ smoothly interpolates between the independent coupling ($\epsilon \to \infty$) and the exact OT coupling ($\epsilon \to 0$), matching the Entropic-CFM panel of Fig. 1, and extends naturally to unpaired settings such as single-cell genomics (Klein et al., 2024).

In summary, the coupling controls how straight and non-crossing the training trajectories are. As we discuss in Section 4.2, this choice turns out to directly affect how well an already trained flow can be steered by inference-time guidance.

2.3 Class-conditional Flow Matching (CCFM)

3 Guidance

3.1 Monte Carlo (MC) estimation

3.2 Simulated-based Monte Carlo estimation

3.3 Covariance Guidance (cov-A / cov-G)

3.4 Learned guidance (GM)

3.4.1 Discussion on guidance methods

The MC method needs only minimal assumptions and minimal tuning in terms of hyperparameter but still produces solid results. However, the MC method does not scale well to higher dimensions because the variance explodes.

The learned (GM) model performs best in terms of W2 distance and energy distance, but several models need to be trained (first disadvantage) and one has to do so for each new energy function (second disadvantage).

The disadvantage of the covariance methods are the additional assumptions that are made, which can be violated in practice.

How does the bias of the guidance methods look like? Maybe something like mode collapse in the Cov methods

3.4.2 Synthetic data experiments

We evaluate the guidance methods on synthetic data and compare the results in terms of W2 distance, energy distance, divergence trace ratio and off-manifold rate. The 2D synthetic data allows to have interpretable results and visualize the generated samples compared to other complex real-world tasks like image generation or single-cell data.

4 Results

4.1 Comparison of guidance methods

We define the energy function J to select three of the eight modes of the target distribution. For this, we define the energy function J as the sum of the squared distances to the three selected modes, scaled by a factor σ^2, which is the energy width.

$$J(x) = -\log \sum_{i \in 0,2,3} \exp\left(-\frac{|x - c_i|_2^2}{2\sigma^2}\right)$$

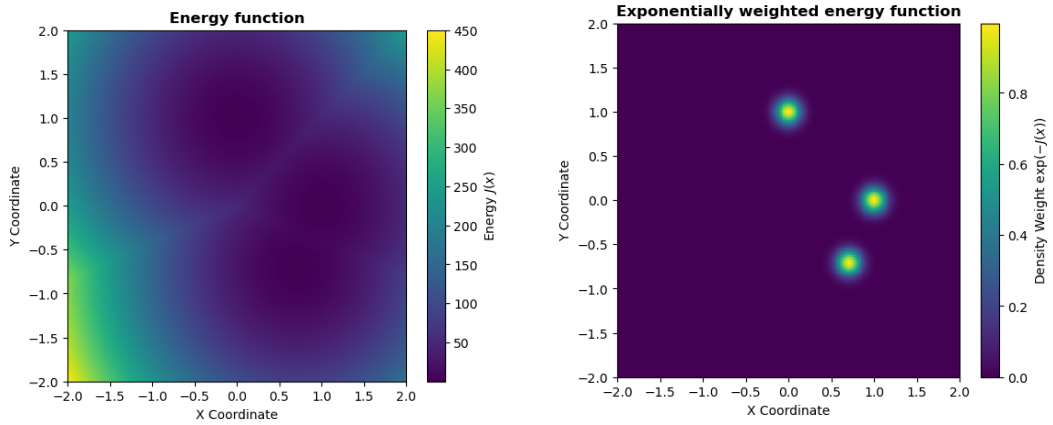
$$J(x) = \frac{|x - c_0|_2^2}{2\sigma^2} + \frac{|x - c_2|_2^2}{2\sigma^2} + \frac{|x - c_3|_2^2}{2\sigma^2}$$

4.2 Limitations of guidance

There are several limitations of guidance methods. Since the MC method is the best inference-steering method, we focus on the limitations of the MC method in this section. In many applications, the OT-coupling is used to learn the velocity field

4.3 Tuning hyperparameters

We want to discuss the effects of the hyperparameters on the guidance methods.



(a) Energy function $J(x)$.

(b) Exponentially weighted energy $e^{-J(x)}$.

Figure 2: Left: the energy function $J(x)$ selecting three of the eight Gaussian modes (low values indicate preferred regions). Right: the corresponding exponentially weighted density $e^{-J(x)}$, proportional to the guided target $p'(x) \propto p_1(x)\, e^{-J(x)}$, with the three target modes clearly localized.

Table 1: Comparison of methods across various metrics. The best method for each metric is **bolded**, and the second best is underlined.

Method	W2 ↓	Energy Dist ↓	Div Trace Ratio	Off-Manifold Rate ↓
Learned (GM)	**0.0802**	**0.0006**	0.9757	0.0659
Class-cond. CFM	0.1632	0.0033	1.0069	0.0786
MC	0.1698	0.0030	**0.9969**	0.0801
cov-G	0.2772	0.0296	1.2113	0.0068
sim-MC	0.3908	0.0739	0.7570	0.5098
cov-A	0.7905	0.2578	0.5410	**0.0008**
CFM	0.8852	0.3804	1.5355	0.6782

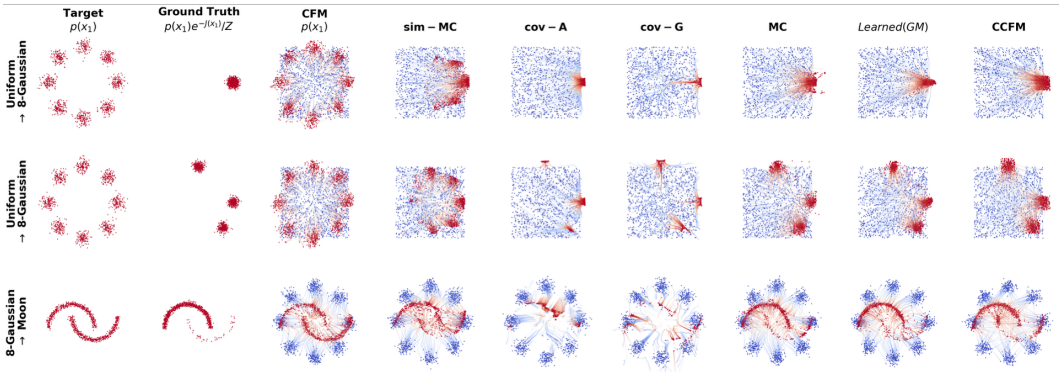


Figure 3: Qualitative comparison of guidance methods across three energy functions (rows). Columns show, from left to right: target distribution, ground truth $p'(x)$, and samples generated by each guidance method (CFM, sim-MC, cov-A, cov-G, MC, Learned (GM), CCFM). Red dots indicate samples in the guided target region; blue dots are the remaining samples.

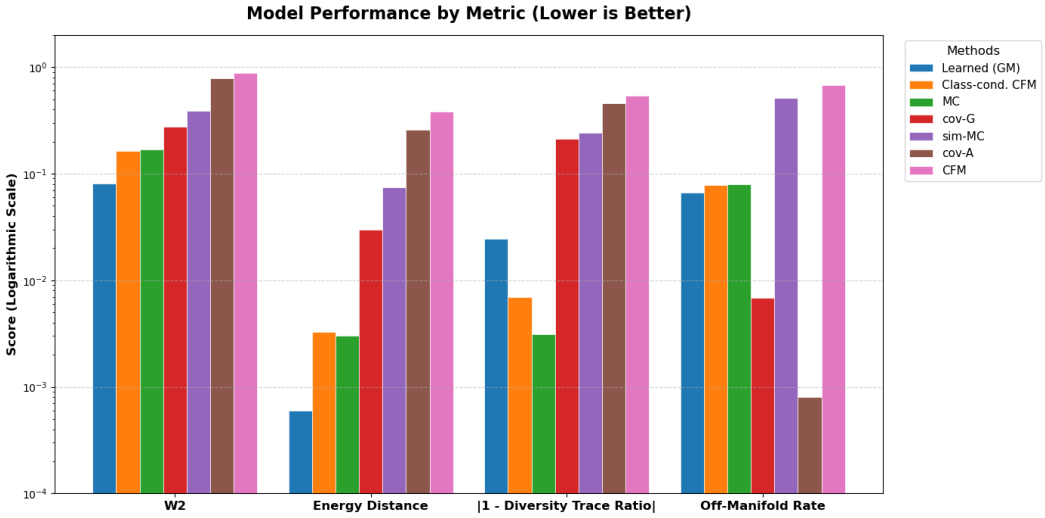
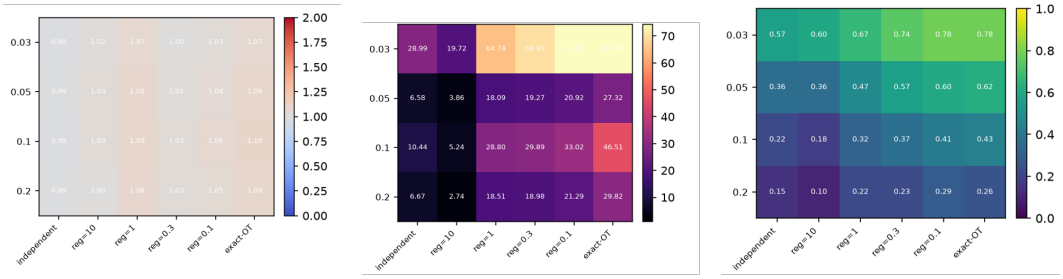


Figure 4: Quantitative comparison of all guidance methods on four metrics shown on a logarithmic scale. Lower is better for all metrics: W2 distance, energy distance, $|1 -$ diversity trace ratio$|$, and off-manifold rate. The learned (GM) model achieves the lowest energy distance; MC guidance offers a strong balance across all metrics without requiring additional training.



(a) Diversity trace ratio. (b) Energy distance. (c) Off-manifold rate.

Figure 5: Phase diagrams for MC guidance across coupling regularization strengths (columns, from independent coupling to exact-OT) and the stabilization parameter ϵ (rows, from 0.03 to 0.2). Stronger coupling (right columns) consistently increases the energy distance and off-manifold rate, indicating that the independence assumption of MC guidance breaks down under tight couplings.

4.3.1 ODE solver

We increased the number of steps in the ODE solver from 100 to 200. This increased the performance of the guidance methods. Moreover, one could use a different ODE solver, e.g. Runge-Kutta 4th order, which might improve the generated samples further with the downside of increased computational cost. An early stopping criterion could be used to mitingate the problem of numerical instabilities at $t = 1$ and mode collapse in the MC guidance method.

4.3.2 MC Guidance

The parameter `mc_ep` (ϵ) appears in the denominator $(1 - t + \epsilon)$ of both the MC importance weights and the velocity correction $u = (x_1 - x_t)/(1 - t + \epsilon)$, acting as a numerical stabilizer as $t \to 1$. Without sufficient ϵ, back-projected source points fall outside the prior support, causing all importance weights to clamp to a uniform distribution and injecting large random forces that scatter particles off the manifold. We tried $\epsilon \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$, with results shown in Table 2. Hence, we increased `mc_ep` from 0.001 to 0.05, which reduced the numerical instability and the outliers of the generated distribution significantly.

Table 2: Effect of the `mc_ep` parameter ϵ on sample quality.

ϵ	W_2 Distance ↓	Off-Manifold Rate ↓
0.001 (default)	0.4244	0.266
0.010	0.3053	0.042
0.050 (tuned)	**0.1091**	**0.002**
0.100	0.2032	0.004

4.3.3 Cov Guidance

5 Conclusion

Conclusion

$p'(x) = \frac{1}{Z} p_1(x) e^{-J(x)}$

Acknowledgments and Disclosure of Funding

Acknowledgement

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A Appendix

Appendix